INCH-POUND
MIL-PRF-1/36F
22 July 1997
SUPERSEDING
MIL-E-1/36E
12 December 1978

PERFORMANCE SPECIFICATION SHEET
ELECTRON TUBES, CATHODE RAY
TYPES 3JP1, 3JP2, 3JP7, 3JP11, AND 3JP12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of specification MIL-PRF-1.

DESCRIPTION: Electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1.

ABSOLUTE RATINGS:

| Parameter: | Type | EF | EC1 | ed | Eb1 | Eb2 | Eb3 | Rg | Zd | Ehk | Eb3/Eb2 | Alt |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | | V | V dc | v | V dc | V dc | V dc | Meg | Meg | V dc | Ratio | ft |
| Maximum: | P1 | 6.9 | 0 | 550 | 1,100 | 2,200 | 4,400 | 1.5 | 1.5 | -125 | 2.3 | 50,000 |
| | P2 | 6.9 | 0 | 550 | 1,100 | 2,200 | 4,400 | 1.5 | 1.5 | -125 | 2.3 | 50,000 |
| | P11 | 6.9 | 0 | 550 | 1,100 | 2,200 | 4,400 | 1.5 | 1.5 | -125 | 2.3 | 50,000 |
| | P7 | 6.9 | 0 | 550 | 1,100 | 2,200 | 4,400 | 1.5 | 1.0 | -125 | 2.3 | 50,000 |
| | P12 | 6.9 | 0 | 550 | 1,100 | 2,200 | 4,400 | 1.5 | 1.0 | -125 | 2.3 | 50,000 |
| Minimum: | All: | 5.7 | -200 | - - - | - - - | 1,500 | 3,000 | - - - | - - - | - - - | - - - | - - - |
| Test Conditions: | | | | | | | | | | | | |
| | P1 | 6.3 | Adj | - - - | Focus | 1,500 | 3,000 | - - - | - - - | - - - | - - - | - - - |
| | P2 | 6.3 | Adj | - - - | Focus | 1,500 | 3,000 | - - - | - - - | - - - | - - - | - - - |
| | P11 | 6.3 | Adj | - - - | Focus | 1,500 | 3,000 | - - - | - - - | - - - | - - - | - - - |
| | P12 | 6.3 | Adj | - - - | Focus | 1,500 | 3,000 | - - - | - - - | - - - | - - - | - - - |
| | P7 | 6.3 | Adj | - - - | Focus | 2,000 | 4,000 | - - - | - - - | - - - | - - - | - - - |

GENERAL:

Qualification - Required


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Table 1. Testing and Inspection.

| Method | Requirement or Test | Type | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| | Qualification inspection | | | | | | |
| 1002 | Barometric pressure, reduced | All | 87.0 mm Hg | | - - - | - - - | - - - |
| 5101 | Neck and bulb alignment (electrostatic types) | All | | dia | - - - | 2.25 | in. |
| 5246 | Focusing voltage, zero bias | P1, P2, P11 | | Eb1 | 300 | 515 | V dc |
| | | | | Eb1 | 400 | 690 | V dc |
| | | P7 | | Eb1 | 302 | 517 | V dc |
| | | P12 | | | | | |
| 5248 | Deflection factor (1D2) | P1, P2, P11 | Eb3= Eb2= 1,500 V dc | DF | 102 | 138 | V dc/in. |
| | | P7 | Eb3= Eb2= 2,000 V dc | DF | 136 | 184 | V dc/in. |
| | | P12 | Eb3= Eb2= 1,500 V dc | DF | 96 | 144 | V dc/in. |
| 5248 | Deflection factor (3D4) | P1, P2, P11 | Eb3= Eb2= 1,500 V dc | DF | 76 | 102 | V dc/in. |
| | | P7 | Eb3= Eb2= 2,000 V dc | DF | 100 | 138 | V dc/in. |
| | | P12 | Eb3= Eb2= 1,500 V dc | DF | 71 | 107 | V dc/in. |
| 5248 | Deflection factor uniformity | All | | - - - | - - - | - - - | - - - |
| 5216 | Cathode illumination | All | | - - - | - - - | - - - | - - - |
| 1331 | Direct-interelectrode capacitance | All | g1 to all | Cg1 | - - - | 10.5 | pF |
| | | | k to all | Ck | - - - | 10.5 | pF |
| | | | D1 to D2 | C1D2 | - - - | 4.0 | pF |
| | | | D3 to D4 | C3D4 | - - - | 3.5 | pF |
| | | | D1 to all | CD1 | - - - | 15.0 | pF |
| | | | D2 to all | CD2 | - - - | 15.0 | pF |
| | | | D3 to all | CD3 | - - - | 12.0 | pF |
| | | | D4 to all | CD4 | - - - | 12.0 | pF |
| 1141 | Pressure (implosion) | All | | - - - | - - - | - - - | - - - |
| 5111 | Vibration | All | | Width | - - - | 1.0 | mm |
| | Quality conformance inspection, part 1 | | | | | | |
| 5201 | Voltage breakdown | All | | - - - | - - - | - - - | - - - |
| 5201 | Voltage breakdown (electrostatic types) | All | | - - - | - - - | - - - | - - - |
| 5206 | Gas “cross” | P1, P2, P11 | Ib3 = 100 μA dc (see note 7) | - - - | - - - | - - - | - - - |
| | | P7 | Ib3 = 200μA dc (see note 7) | - - - | - - - | - - - | - - - |
| | | P12 | Ib3 = 30μA dc (see note 7) | - - - | - - - | - - - | - - - |
| 5106 | Bulb, screen, and faceplate quality | All | | - - - | - - - | - - - | - - - |
| - - - | Deflection plate cutoff | P1, P2, P7,P11 | Ib3 = 10μA dc (max)(see note 3) | - - - | - - - | - - - | - - - |
| | | P12 | Ib3 = 5μA dc(max) (see note 3 and 4) | - - - | - - - | - - - | - - - |



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Table 1. Testing and Inspection -Continued.

| Method | Requirement or Test | Type | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 1 - continued | | | | | | |
| 5223 | Modulation | P7<br>P12 | Ib3 = 200 µA dc<br>Ib3 = 100 µA dc | ΔEc<br>ΔEc | | 55<br>40 | V dc<br>V dc |
| 5231 | Spot position (electrostatic deflection) | P1, P2<br>P11,P12<br>P7 | | - - -<br><br>- - - | - - -<br><br>- - - | 15<br><br>12 | mm<br><br>mm |
| 5231 | Spot displacement (leakage) | All | | Displ | - - - | 7 | mm |
| 5241 | Grid-cutoff voltage | P1, P2<br>P11,P12<br>P7 | | Ecl<br><br>Ecl | -22.5<br><br>-30 | -67.5<br><br>-90 | V dc<br><br>V dc |
| 5251 | Grid No. 1 leakage current | All | | - - - | - - - | - - - | - - - |
| 5251 | Anode No. 2 leakage current | All | | - - - | - - - | - - - | - - - |
| 5221 | Light output | P1<br><br>P11 | Eb2 = 1,500 V; Eb3 = 3,000 V; Ib3 = 100 µA dc<br>Eb2 = 1,500 V; Eb3 = 3,000 V; Ib3 = 100 µA dc (see note 5) | Light<br><br>Light | 25<br><br>9 | - - -<br><br>- - - | ftL<br><br>ftL |
| | Quality conformance inspection, part 2 | | | | | | |
| 1301 | Heater current | All | | If | 540 | 660 | mA |
| 5201 | Electrode current (anode No. 1) | P1, P2, P11<br>P7<br>P12 | Ib3 = 100 µA dc<br><br>Ib3 = 200 µA dc<br>Ib3 = 5 µA dc (see note 1) | Ib1<br><br>Ib1<br>Ib1 | -50<br><br>-50<br>-50 | 10<br><br>10<br>10 | µA dc<br><br>µA dc<br>µA dc |
| 5101 | Base alignment (electrostatic types) | All | +1D2, pin No. 5 | - - - | - - - | - - - | - - - |
| 5101 | Side terminal alignment (electrostatic types) | All | +1D2 | - - - | - - - | - - - | - - - |
| 5201 | Electrode current (anode No. 3) | P1, P2, P11 | See note 2 | Ib3 | 150 | - - - | µA dc |
| 5201 | Electrode current (cathode) | P1, P2, P11<br>P7<br>P12 | Ib3 = 100 µA dc<br><br>Ib3 = 200 µA dc<br>Ib3 = 5 µA dc (see note 1) | Ik<br><br>Ik<br>Ik | - - -<br><br>- - -<br>- - - | 1,000<br><br>1,500<br>100 | µA dc<br><br>µA dc<br>µA dc |



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Table 1. Testing and Inspection -Continued.

| Method | Requirement or Test | Type | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 2 - Continued | | | | | | |
| 5101 | Angle between traces | All | | - - - | - - - | - - - | - - - |
| 5101 | Neck and base alignment (electrostatic types) | All | | - - - | - - - | - - - | - - - |
| 5216 | Stray light emission (conventional types) | All | Eb2 = 2,200 V dc; Eb3 = 4,400 V dc | - - - | - - - | - - - | - - - |
| 5221 | Screens | P2 | | D1(1) | 360 | - - - | cB |
| | | P7 | | D1(5) | 400 | - - - | cB |
| | | | | G5 : 1 | 4 | - - - | |
| | | P12 | | cbf | | | |
| | | | | D0.1 (E) | 550 | - - - | - - - |
| | | | | D0.3 (E) | 80 | 100 | cB |
| 5226 | Line width “A” (electrostatic deflection) | P1, P2, P11 | Ib3 = 100 μA dc (see note 6) | Width | - - - | 0.75 | mm |
| | | P7 | Ib3 = 200 μA dc (see note 6) | Width | - - - | 0.80 | mm |
| | | P12 | Ib3 = 5 μA dc (see notes 1 and 6) | Width | - - - | 0.35 | mm |
| 5226 | Line width “B” (electrostatic deflection) | P1, P2, P11 | Ib3 = 100 μA dc (see note 8) | Width | - - - | 0.90 | mm |
| | | P7 | Ib3 = 200 μA dc (see note 8) | Width | - - - | 1.00 | mm |
| | | P12 | Ib3 = 5 μA dc (see notes 1 and 8) | Width | - - - | 0.40 | mm |
| 5246 | Focusing voltage at cutoff | P1, P2, P11 | | Eb1 | 345 | 515 | V dc |
| | | P7 | | Eb1 | 460 | 690 | V dc |
| | | P12 | | Eb1 | 345 | 517 | V dc |
| 5248 | Deflection factor (1D2) | P1, P2, P11,P12 | | DF | 127 | 173 | V dc/in. |
| | | P7 | | DF | 170 | 230 | V dc/in. |
| 5248 | Deflection factor (3D4) | P1, P2, P11,P12 | | DF | 94 | 128 | V dc/in. |
| | | P7 | | DF | 125 | 170 | V dc/in |
| 5251 | Heater-cathode leakage current | All | | - - - | - - - | - - - | - - - |
| 1101 | Secureness of base, cap, or insert | Al | | - - - | - - - | - - - | - - - |
| 1105 | Permanence of marking | | | - - - | - - - | - - - | - - - |
| 1111 | Base pin solder depth | | | - - - | - - - | - - - | - - - |



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Table 1. Testing and Inspection -Continued.

| Method | Requirement or Test | Type | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 3 | | | | | | |
| - - - | Life-test provisions | P1, P2, P7, P11 | Group C; t = 500 hours (min)<br>Eb3 = 4,400 V dc;<br>Eb2 = 2,200 V dc;<br>Ib3 = 30 μA dc | - - - | - - - | - - - | - - - |
| | | P12 | Group C; t = 500 hours (min)<br>Eb3 = 4,400 V dc<br>Eb2 = 2,200 V dc<br>Ib3 = 5 μA dc | - - - | - - - | - - - | - - - |
| - - - | Life-test end points: | | | | | | |
| 5223 | Modulation | P1, P2, P11,P12 | Ib3 = 100 μA dc | ΔEc | - - - | 40 | V dc |
| | | P7 | Ib3 = 150 μA dc | ΔEc | - - - | 55 | V dc |
| 5226 | Line width “A” (electrostatic deflection) | P1, P2, P11 | Ib3 = 100 μA dc | Width | - - - | 0.75 | mm |
| | | P7 | Ib3 = 150 μA dc | Width | - - - | 0.86 | mm |
| | | P12 | Ib3 = 5 μA dc | Width | - - - | 0.35 | mm |
| 5226 | Line width “B” (electrostatic deflection) | P1, P2, P11 | Ib3 = 100 μA dc | Width | - - - | 0.90 | mm |
| | | P7 | Ib3 = 150 μA dc | Width | - - - | 1.12 | mm |
| | | P12 | Ib3 = 5 μA dc | Width | - - - | 0.40 | mm |
| 1331 | Direct-interelectrode capacitance | All | g1 to all | Cgl | - - - | 10.5 | pF |
| | | | k to all | Ck | - - - | 10.5 | pF |
| | | | D1 TO D2 | C1D2 | - - - | 4.0 | pF |
| | | | D3 TO D4 | C3D4 | - - - | 3.5 | pF |
| | | | D1 TO ALL | CD1 | - - - | 15.0 | pF |
| | | | D2 TO ALL | CD2 | - - - | 15.0 | pF |
| | | | D3 TO ALL | CD3 | - - - | 12.0 | pF |
| | | | D4 TO ALL | CD4 | - - - | 12.0 | pF |
| | | | (See Note 10) | | | | |

NOTES:

1 / The tube can be severely and permanently damaged if the current density on the screen is allowed to rise too high in static tests of this sort. For this reason, the length of time during which the screen is bombarded should be kept as short as possible compatible with the reading of specified currents limits.

2 / Ib3 is to be measured with ΔEc = 40 V dc when cutoff is greater than -40 V dc and with Ecl = 0 when cutoff is less than -40 V dc.

3 / At specified conditions there shall be no pattern cutoff with the useful screen area.

4/ To prevent burning of the screen, the beam current should not be allowed to exceed that amount necessary to just produce a visible trace.



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NOTES - Continued:

5/ As measured by a 2 x 2 inch (50.80 x 50.80 mm ) raster using a type 3 photronic cell without eye correction, calibrated in foot candles of illumination from a light source having color temperature of 2,700°K.

6/ The reflecting plates shall be returned to anode No. 2 through a 2.5 megohm resistance. Ib3, the beam current, shall be set as follows:

P1- - - - - - - - - Ib3 = 100 μA dc
P2 - - - - - - - - - Ib3 = 100 μA dc
P7- - - - - - - - - Ib3 = 200 μA dc
P11 - - - - - - - - - Ib3 = 100 μA dc
P12 - - - - - - - - - Ib3 = 5 μA dc

a/ The high frequency scanning shall be applied to the deflecting plates nearest the screen and the amplitude shall be adjusted to give a line length of approximately 90 percent of the maximum tube diameter. The low-frequency scanning amplitude shall be expanded to approximately 90 percent of the maximum tube diameter in the direction perpendicular to the direction of high-frequency scanning. Readjustment may be made for best overall focus. The tube shall be observed for deflection defocusing, astigmatism, or spot ellipticity observable to the eye as evidenced by fuzziness due to lack of sharpness of the trace (usually around the edges), bow-tying (irregular widths of any single line when observed at different points), bowing of trace other than that normally caused by curvature of bulb.

b/ This test for focus is to be made in addition to the line width measurements.

7/ This test to be performed at the conclusion of the holding period.

8/ The same conditions shall be set up as described in note 6 except that the connection of deflection elements to the low- and high-frequency scanning supplies shall be interchanged and the amplitudes adjusted to 90 percent of the maximum tube diameter in both directions without any adjustment of focus from note 6 conditions.

An examination for defocusing, astigmatism, or spot ellipticity shall be made as in note 6.

9/ Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

10/ This test shall be performed during the initial production and once each succeeding 12- calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. The regular ‘12-calender month” double sampling plan shall be reinstated after three consecutive samples have been accepted.



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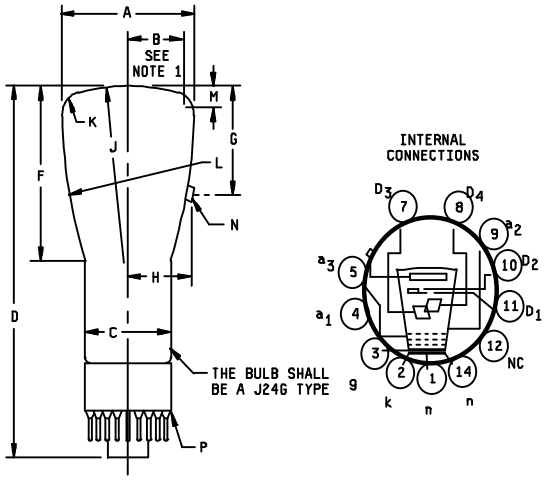


| Dimensions in inches with metric equivalents (mm) in parentheses (see note b) | | |
|---|---|---|
| Ltr | Minimum | Maximum |
| Quality conformance inspection, part 2 | | |
| A | 2.94 (74.7) | 3.06 (77.7) |
| B | 1.37 (34.8) | |
| C | 1.94 (49.3) | 2.06 (52.3) |
| D | 9.75 (247.6) | 10.25 (260.4) |
| F | 3.75 (95.2) | 4.00 (101.6) |
| G | 1.50 (38.1) | 2.00 (50.8) |

| Table 1 continued. | |
|---|---|
| Quality conformance inspection, part 3 (See note 10) | |
| N | Bulb contact: J1-22 |
| P | Base: B12-37 |
| Reference dimensions (see note c) | |
| H | 1.62 (41.1) |
| J | 8.00 (203.2) R |
| K | .25 ( 6.4) R |
| L | 12.44 (316.0) R |
| M | .35 ( 8.9) |

NOTES:
a/ The minimum useful screen radius shall not be less than 1.375 inches (34.92 mm).
b/ Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
c/ Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 3JP1, 3JP2, 3JP7, 3JP11, AND 3JP12.



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Navy - EC
Air Force - 85

Review activities:
Army - AR, AV, ME
Navy - AS, CG, OS
Air Force - 11, 99

Preparing activity:
DLA - CC

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